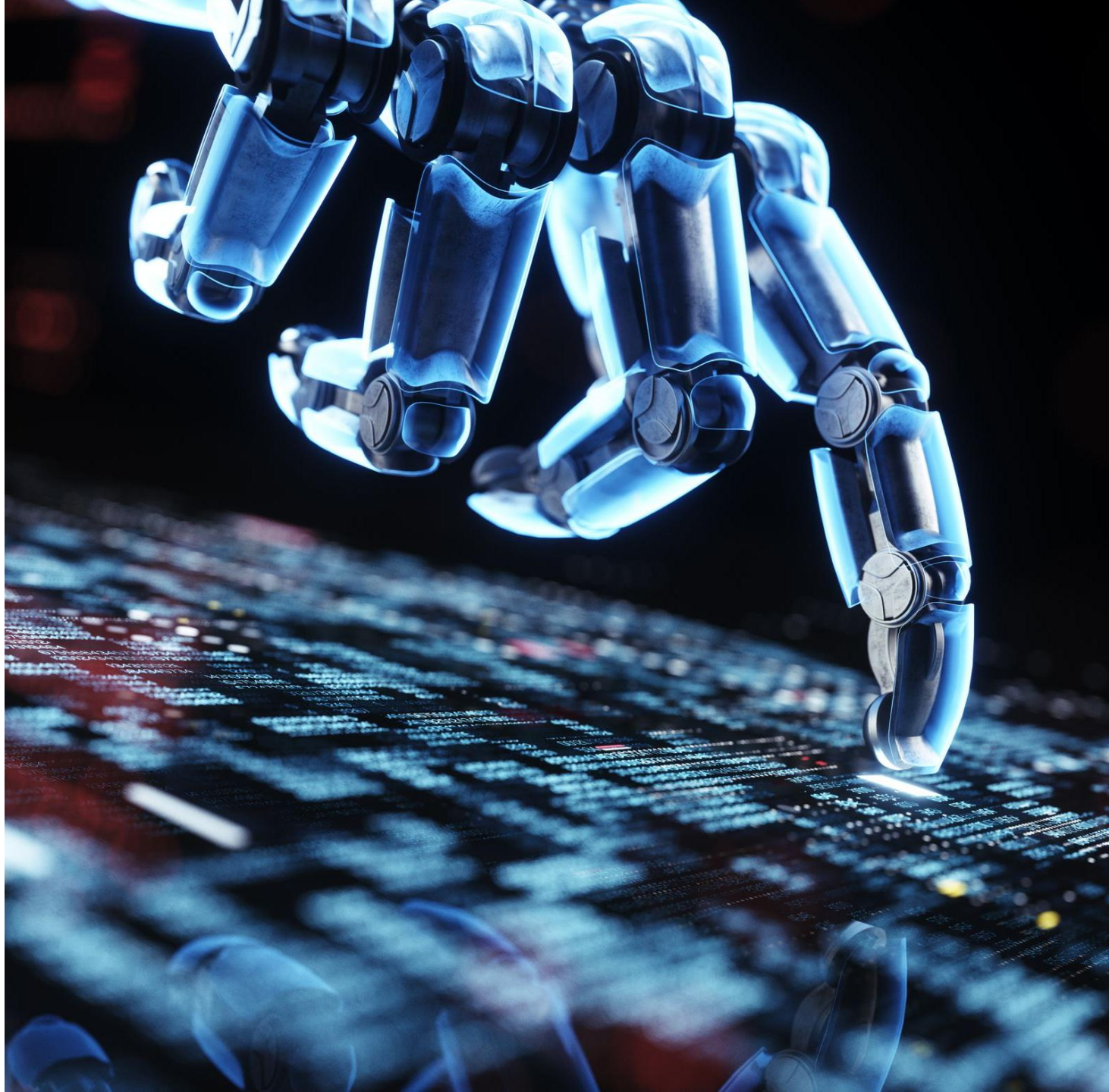


AI Foundations and Machine Learning Fundamentals

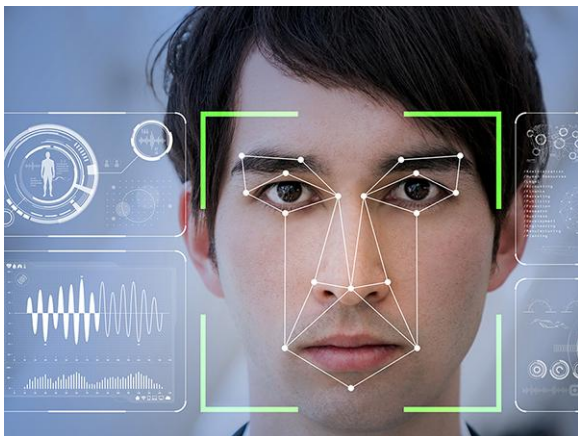




Introduction to AI

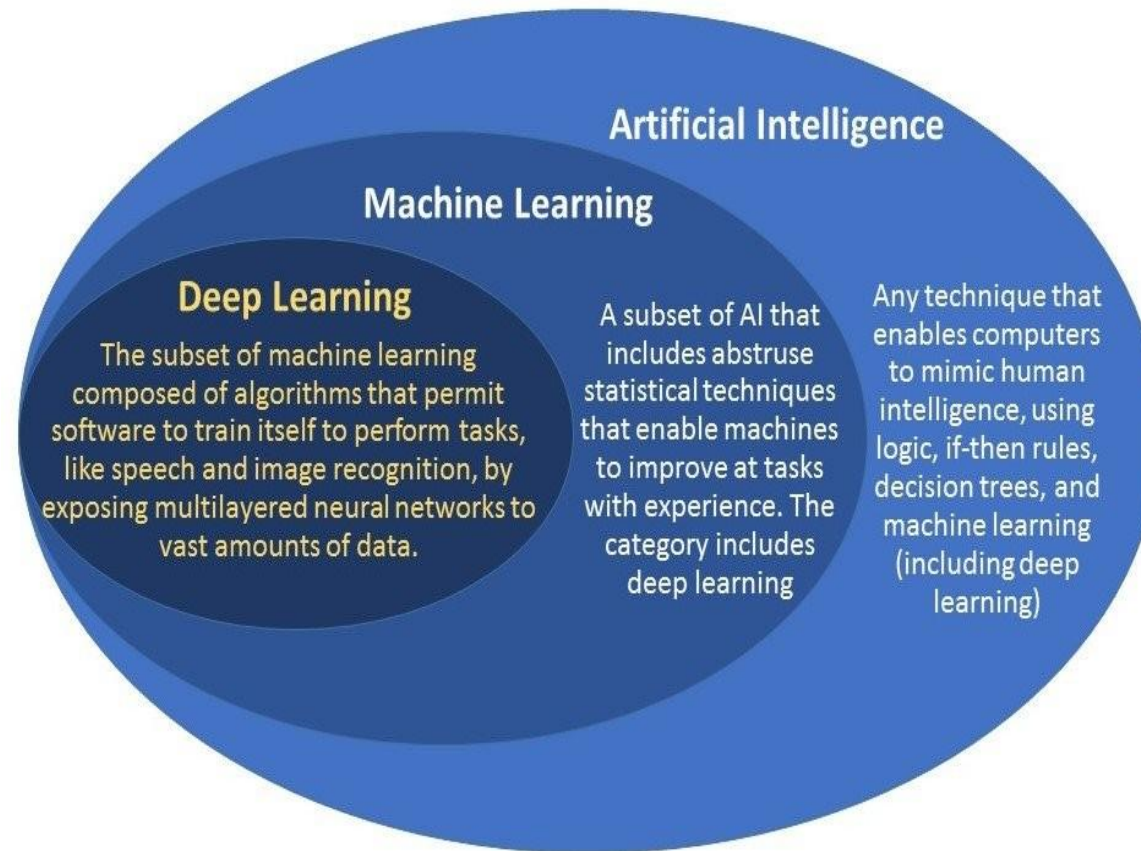
- **Artificial Intelligence** refers to computer systems that can perform complex tasks normally done by human-reasoning, decision making, creating, etc.
- There is **no single, simple definition** of **artificial intelligence** because AI tools are capable of a wide range of tasks and outputs. ([Link:NASA](#))
- Important Points:
 - Any **artificial system** that performs tasks under varying and unpredictable circumstances without significant human oversight, or that can learn from experience and improve performance when exposed to data sets. E.g **autonomous vehicle (self-driving car)**.
 - An **artificial system** developed in computer software, physical hardware, or other context that solves tasks requiring **human-like perception, cognition, planning, learning, communication, or physical action**. E.g **Humanoid Robots like Sophia** (developed by Hanson Robotics)
 - An **artificial system** designed to **think** or **act** like a **human**, including cognitive architectures and neural networks.
 - A set of techniques, including **machine learning** that is designed to approximate a cognitive task.
 - An **artificial system** designed to **act rationally**, including an intelligent **software agent** or embodied **robot** that **achieves goals** using **perception, planning, reasoning, learning, communicating, decision-making, and acting**.

Examples



IBM Watson Health

Evolution of AI

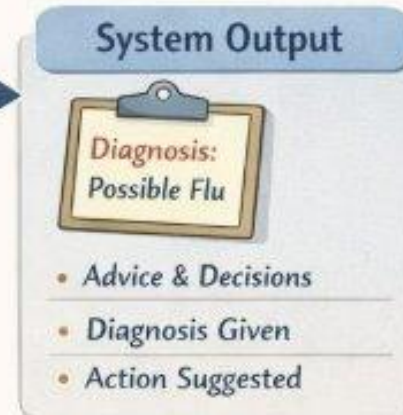
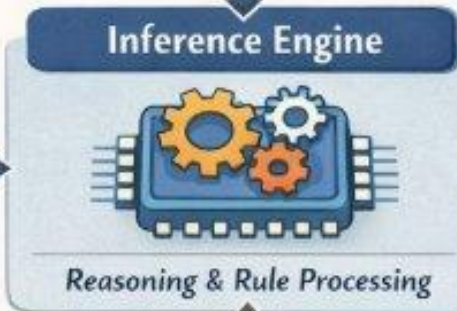
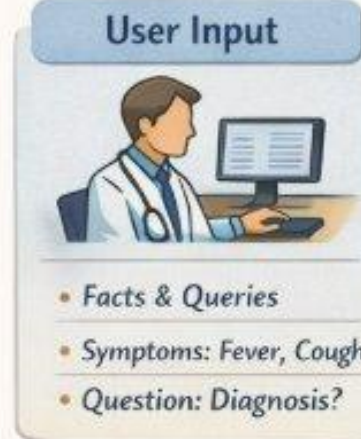


Rule-Based Systems in Early AI

Expert System Architecture

Key Features of Rule-Based Systems

- Built on IF-THEN rules crafted by human experts
- Simple, transparent, and easy to understand
- Great for diagnostics, decision support, and automation
- Ideal for domains like medical, finance, & industrial





AI vs ML vs DL vs Generative AI



1 AI

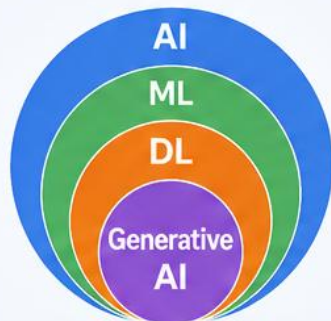


A broad field focused on making machines intelligent.



Key Point

AI is the umbrella that includes ML, DL, and Generative AI.



2 Machine Learning (ML)



Machines learn patterns from data.



Example



Predicting house prices

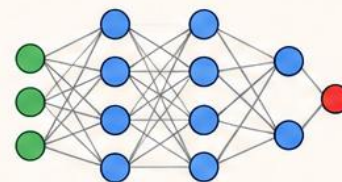


Spam detection



Product recommendation

3 Deep Learning (DL)



Uses neural networks with multiple layers.



Example



Face Recognition



Speech Recognition



Medical Image Analysis

4 Generative AI



Creates new content.



Example



ChatGPT



Gemini



Claude



Midjourney



In Simple Terms:



AI is the goal



ML learns from data



DL learns deeper with layers

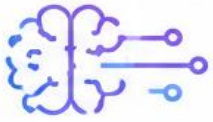


Generative AI creates something new



Why has AI become popular now?

- More data
- Better GPUs
- Cloud Computing
- Better Algorithms
- Transformers



AI Applications Across Industries



AI is transforming every industry by solving real-world problems and creating value.



Healthcare



Disease Detection



Medical Imaging



Drug Discovery



Finance



Fraud Detection



Credit Risk Assessment



Algorithmic Trading



Manufacturing



Predictive Maintenance



Quality Inspection



Education



Personalized Learning



AI Tutors



Automated Assessment



Retail



Recommendations



Demand Forecasting



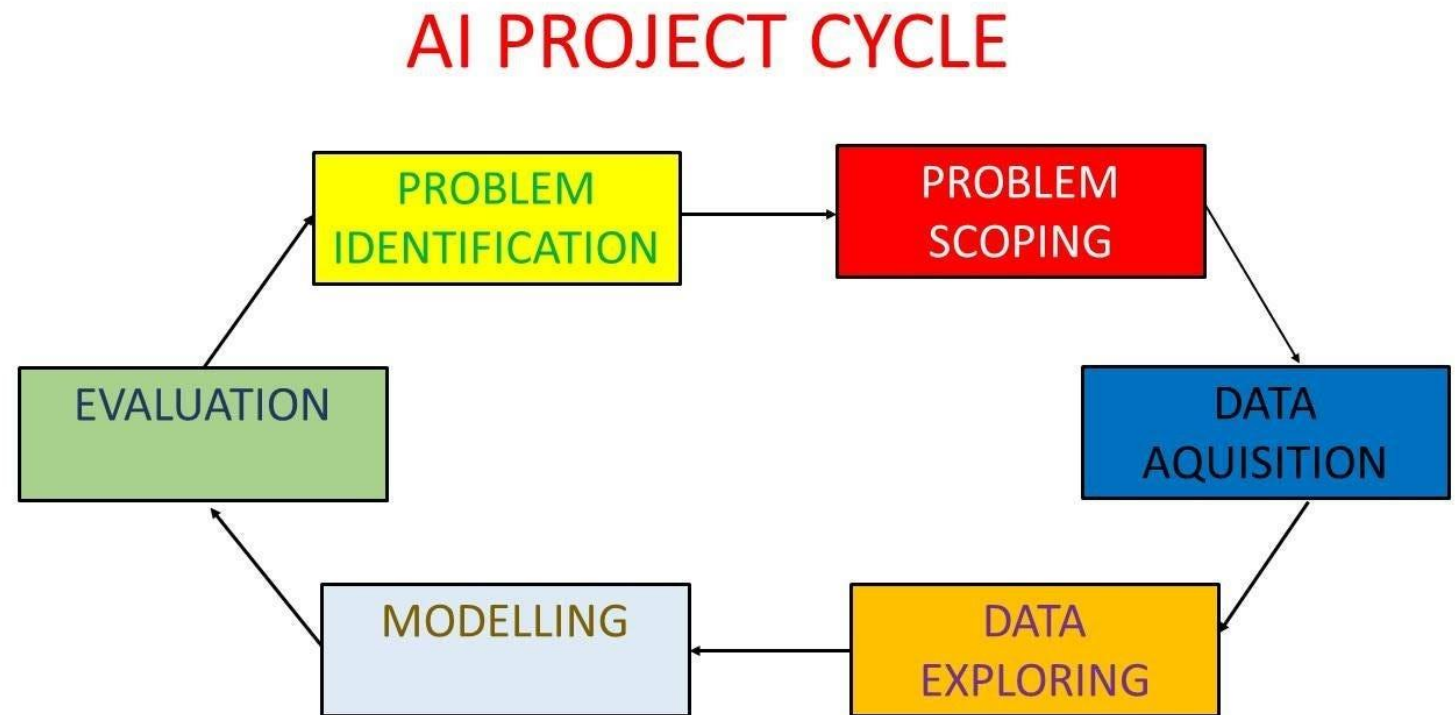
Customer Support



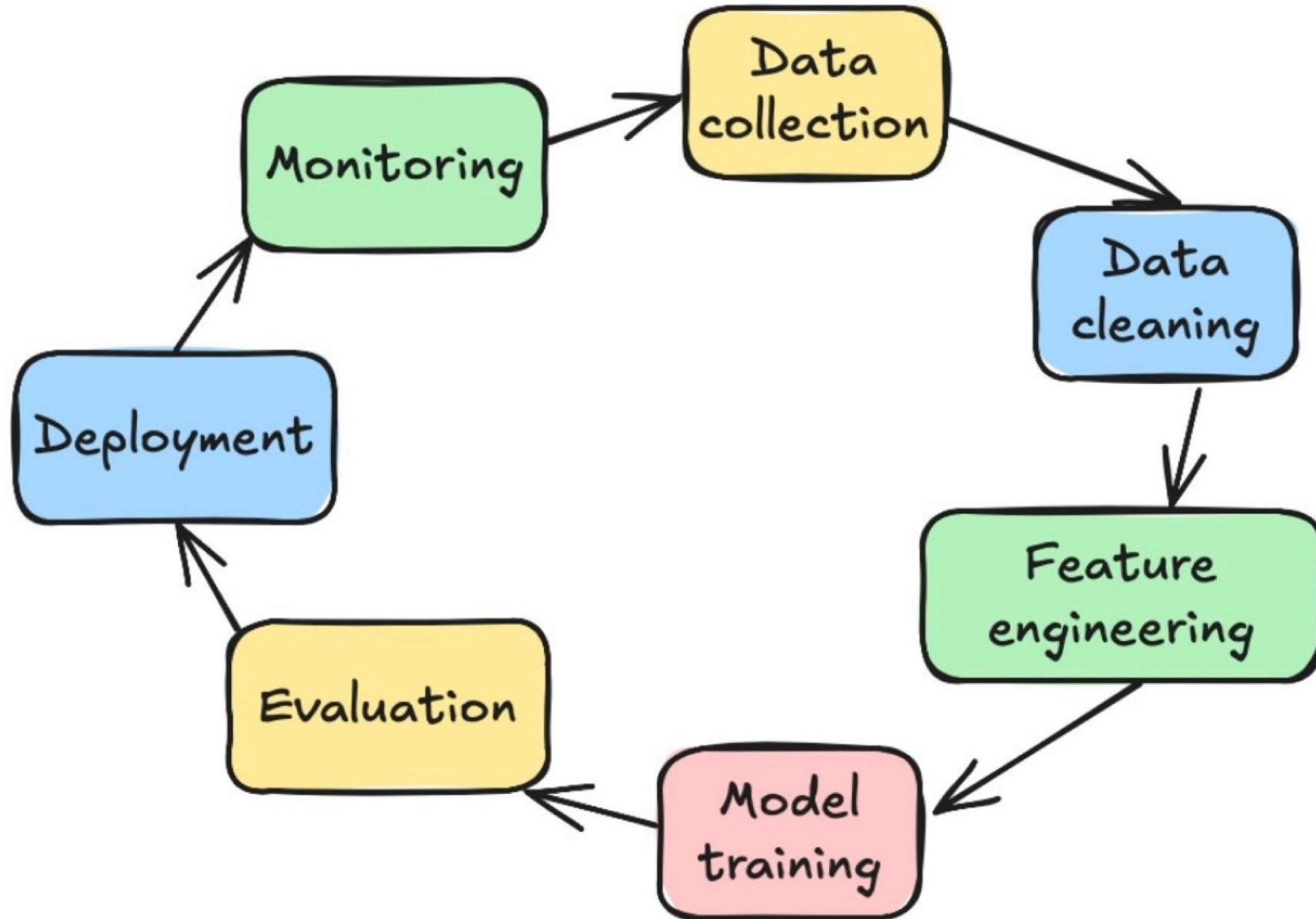
Smarter Decisions. Higher Efficiency. Better Experiences. That's the **Power of AI.**



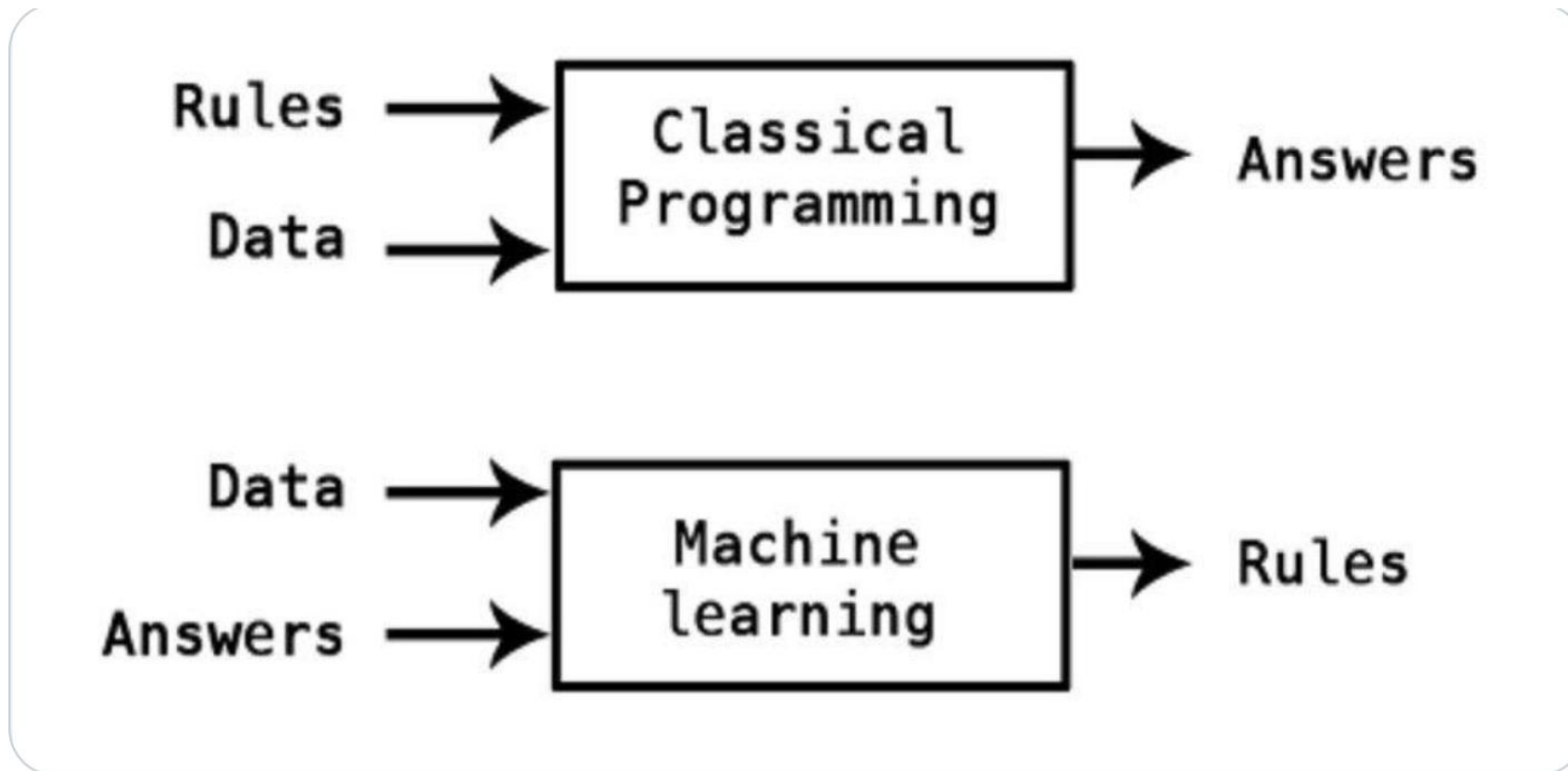
AI Project Cycle



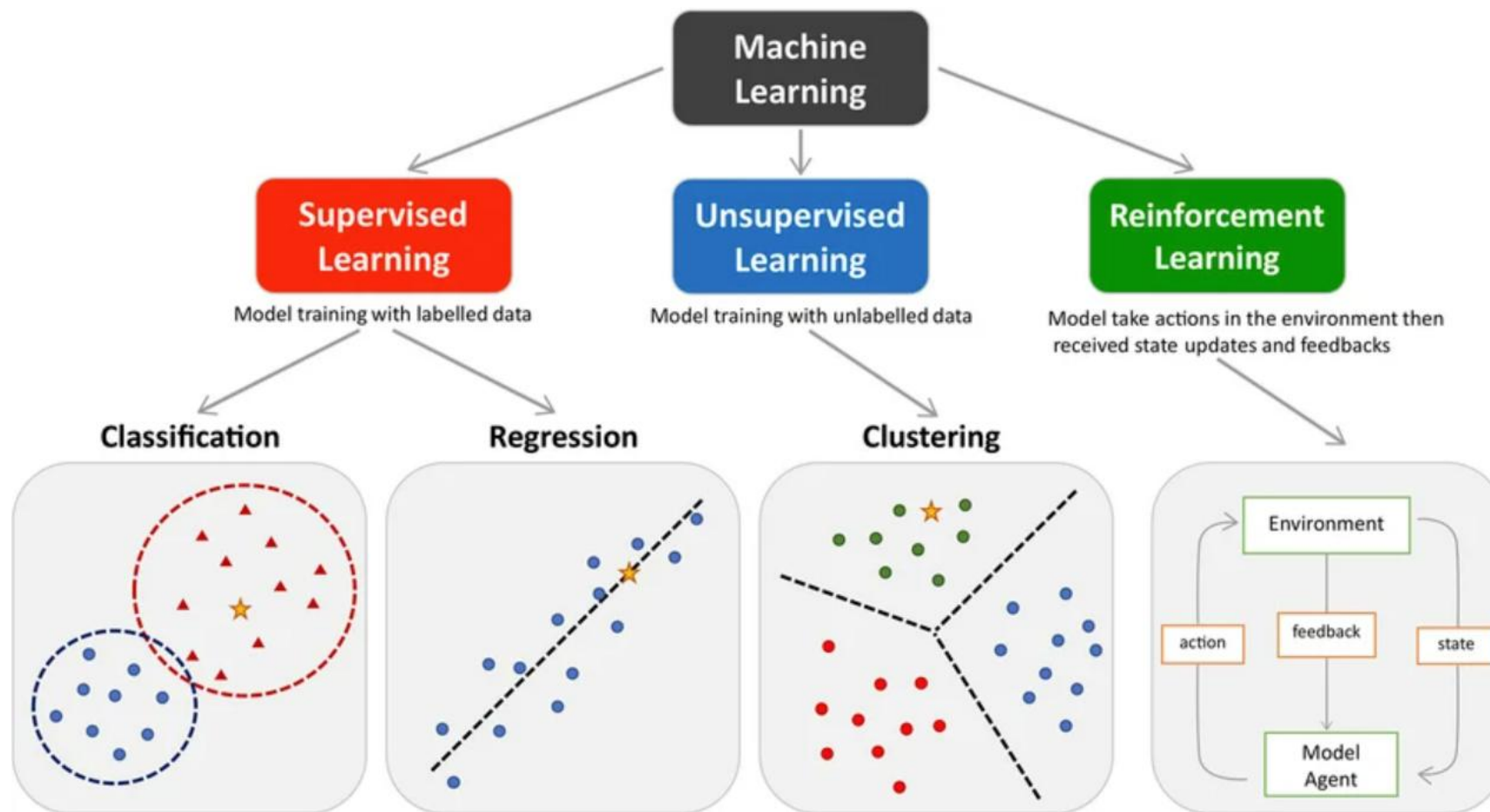
Machine Learning Cycle



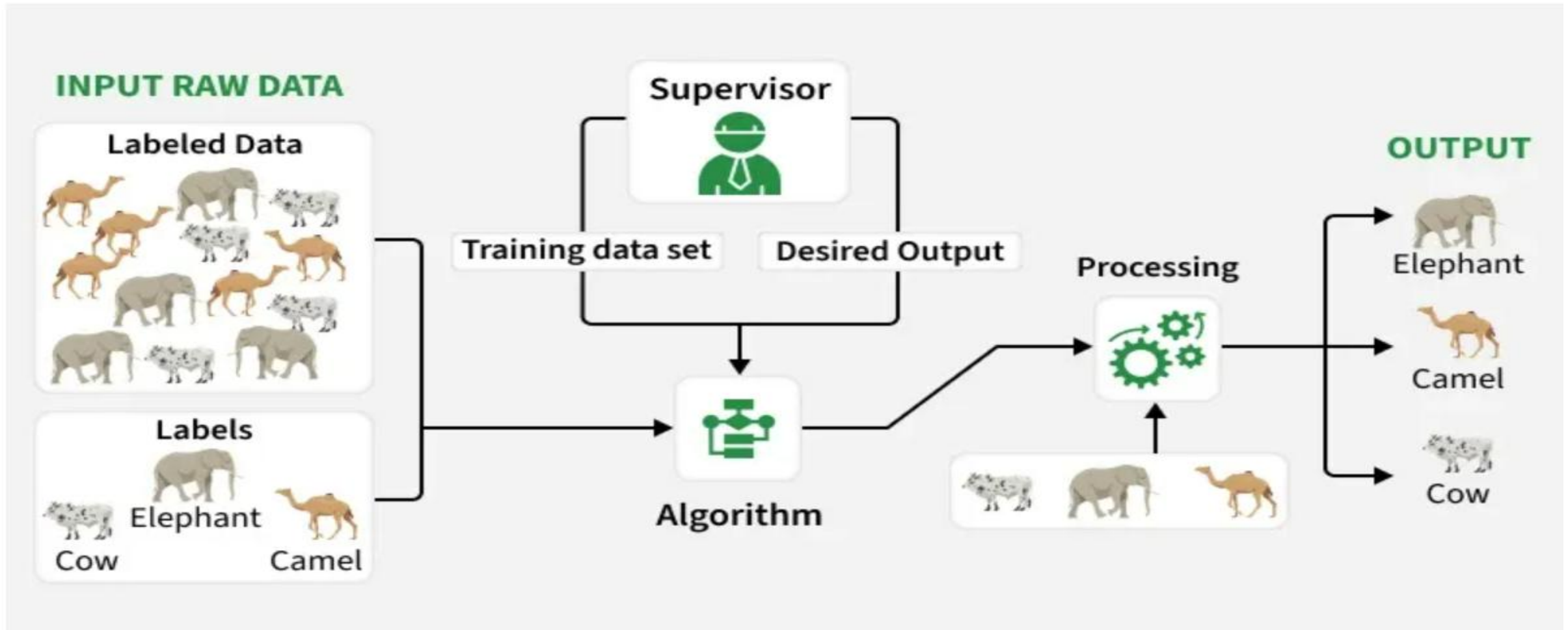
Rule-based vs Machine Learning



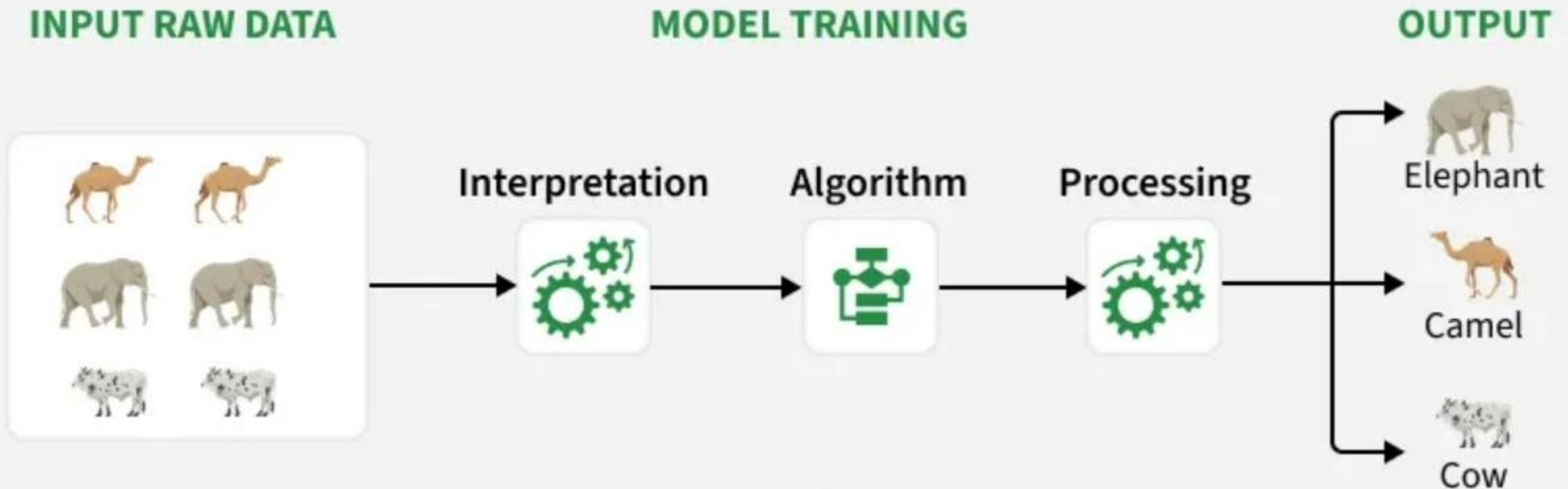
Types of Machine Learning



Supervised Machine Learning

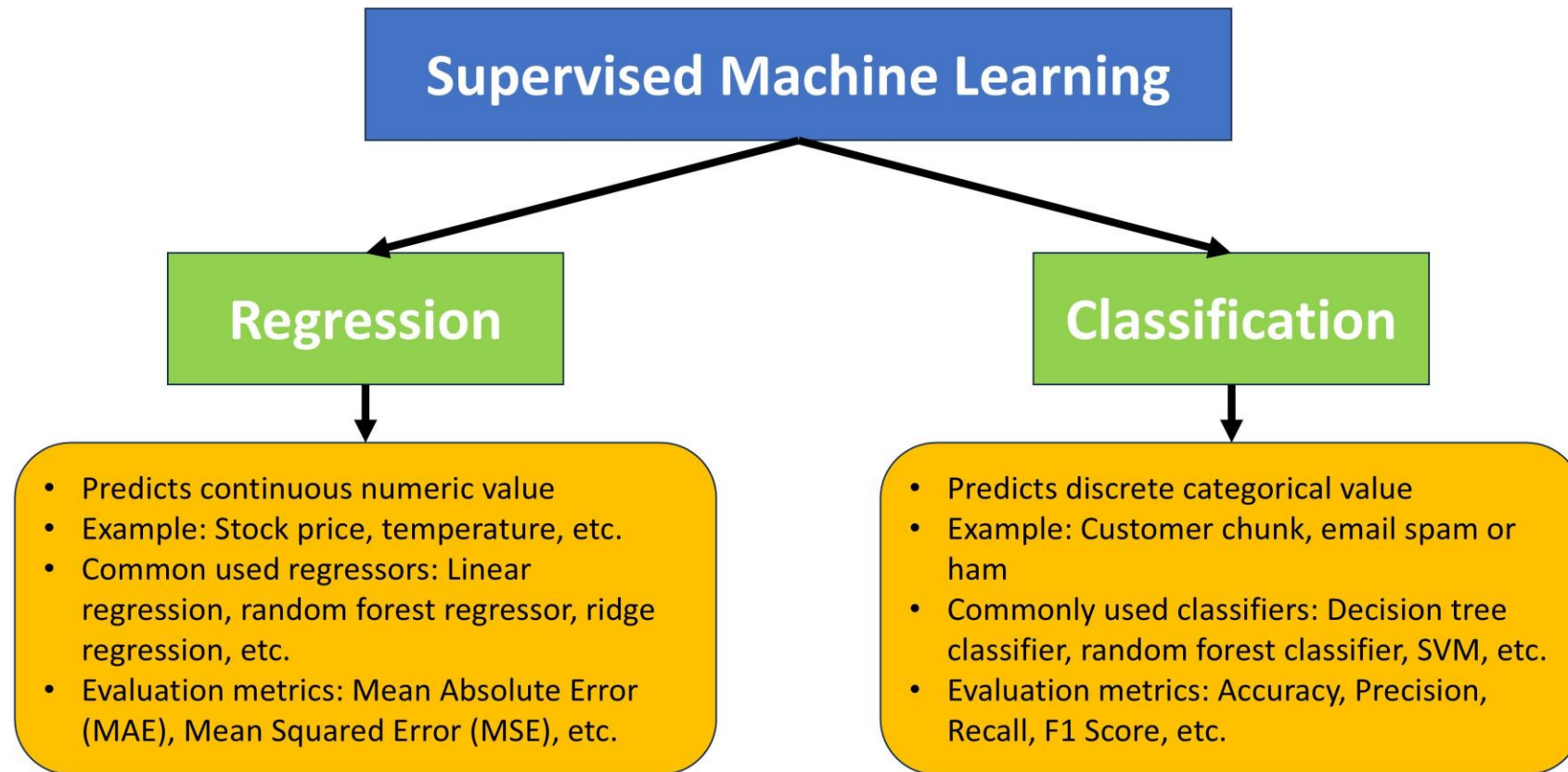


Un-Supervised Machine Learning



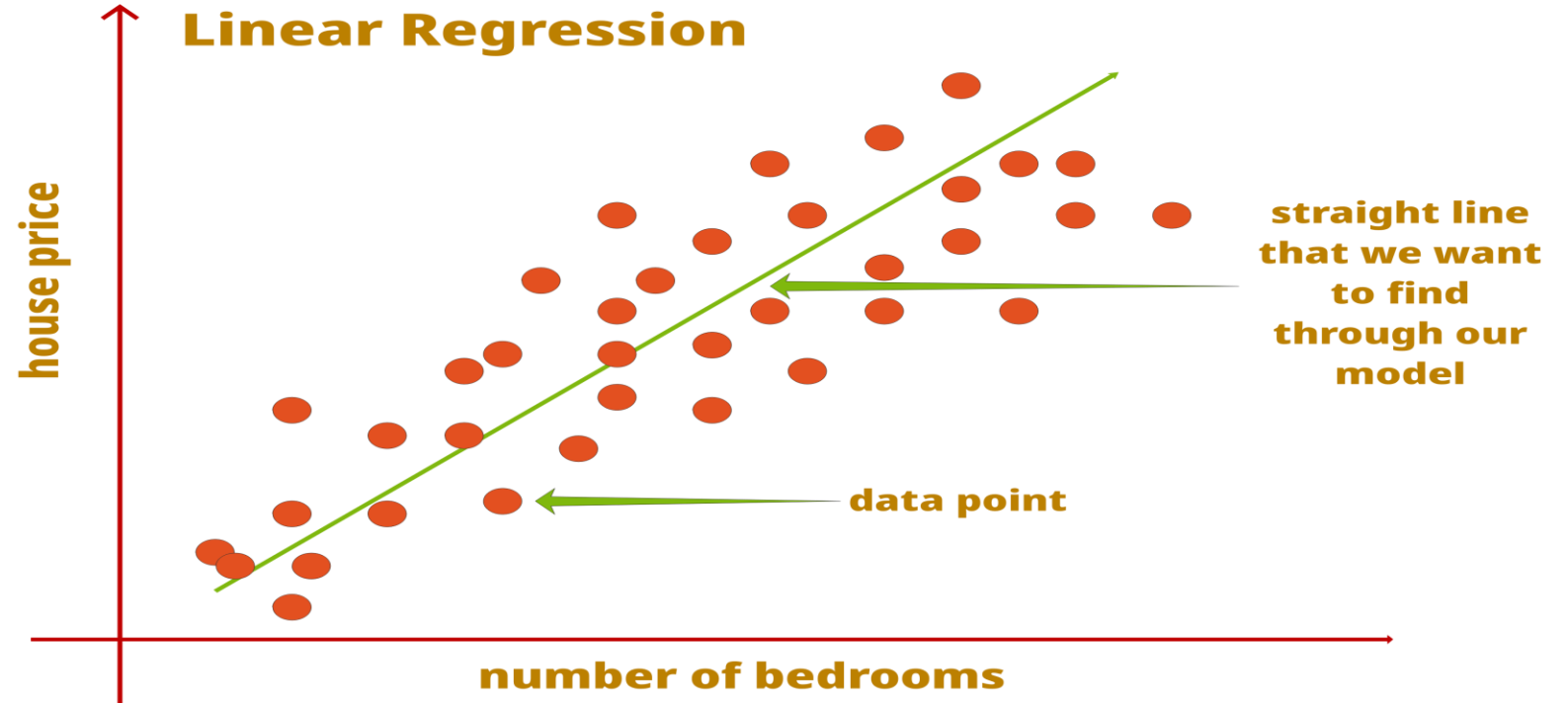
- Unknown Output
- No Training Data Set

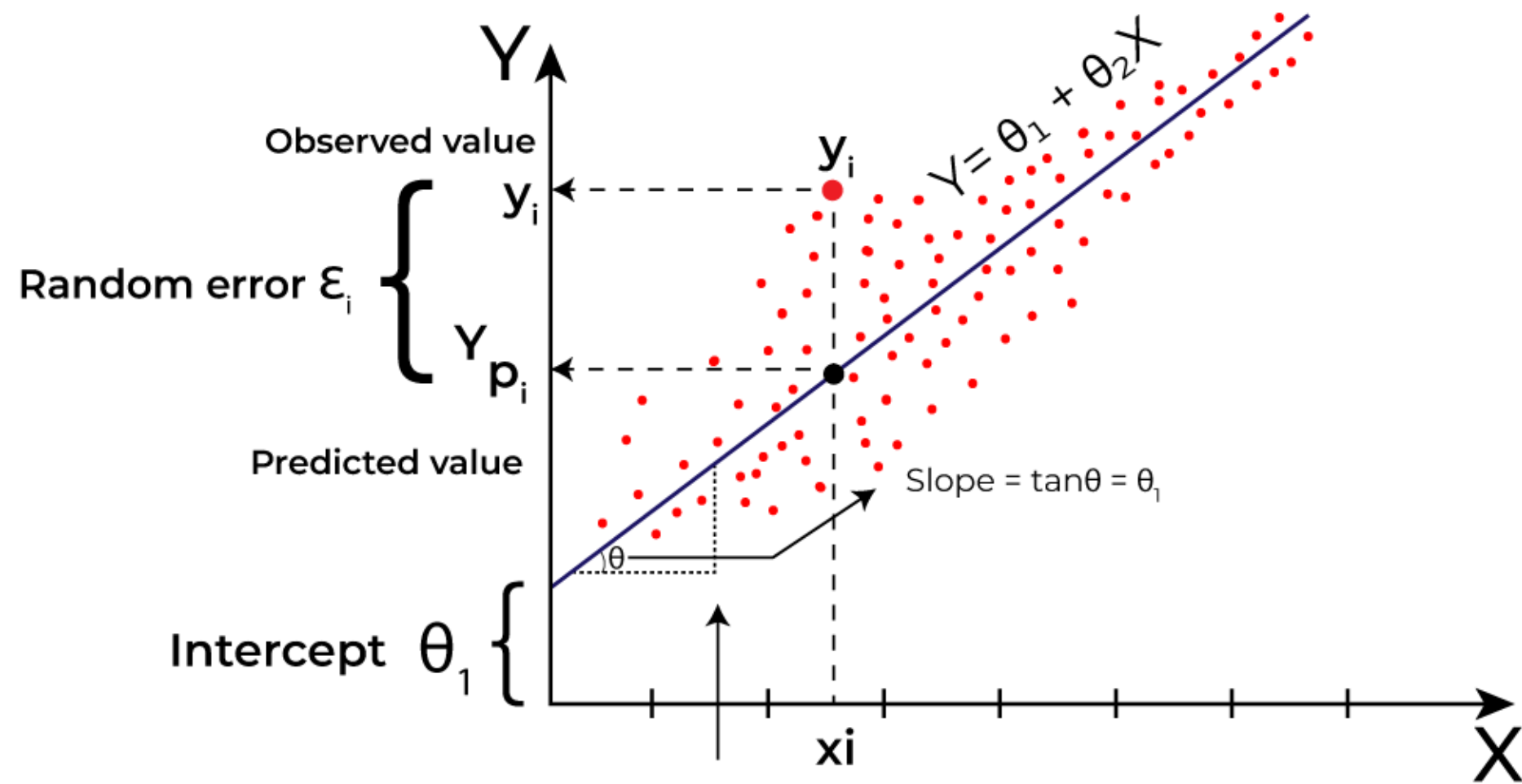
Regression Vs Classification Analysis



Linear Regression

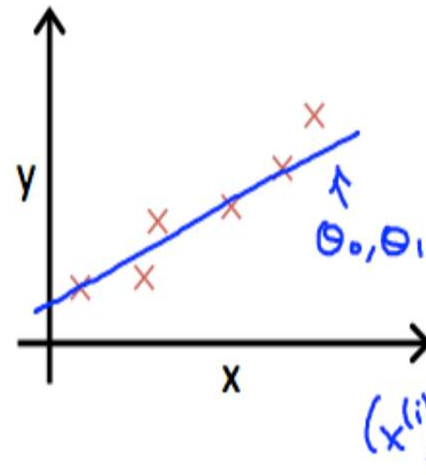
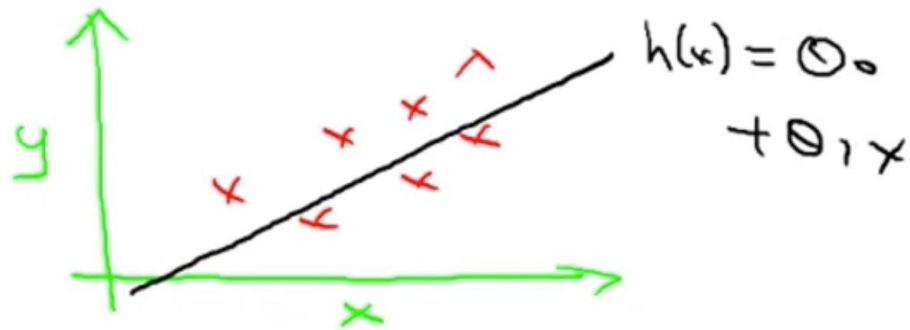
1. Linear regression is a supervised machine learning and statistical algorithm used to model the relationship between a dependent variable (y) and one or more independent variables (x).
2. It predicts continuous outcomes by fitting the best straight line (or hyperplane) that minimizes the gap between actual(y) and predicted values(h - $\theta(x)$).





$$h_{\theta}(x) = \theta_0 + \theta_1 x$$

Shorthand: $h(x)$



Idea: Choose θ_0, θ_1 so that $h_{\theta}(x)$ is close to y for our training examples (x, y)

minimize θ_0, θ_1

$\frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$

#training examples

$h_{\theta}(x^{(i)}) = \theta_0 + \theta_1 x^{(i)}$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

minimize $J(\theta_0, \theta_1)$

θ_0, θ_1

Cost function

Squared error function

Gradient descent algorithm

repeat until convergence {

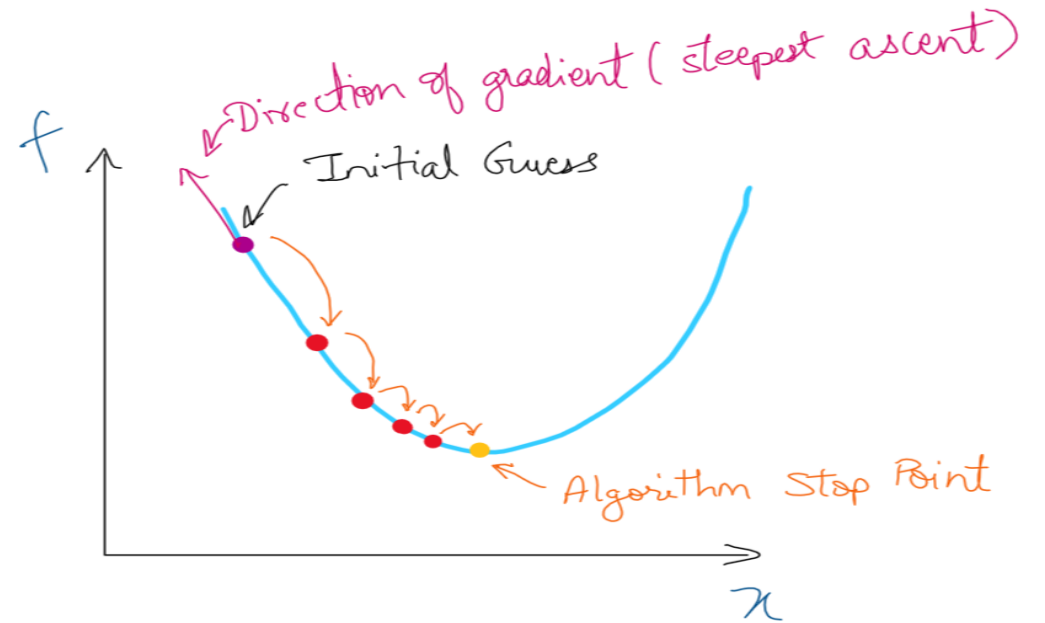
$$\theta_0 := \theta_0 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})$$
$$\theta_1 := \theta_1 - \alpha \frac{1}{m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

}

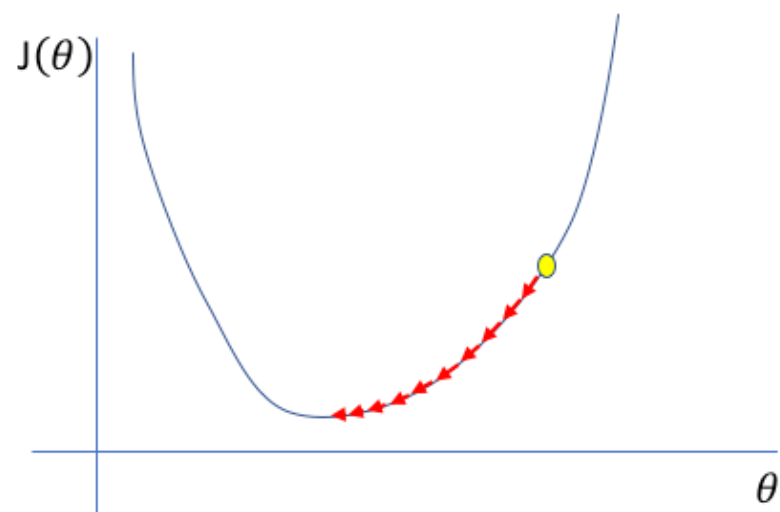
update θ_0 and θ_1 simultaneously

$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$

$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$

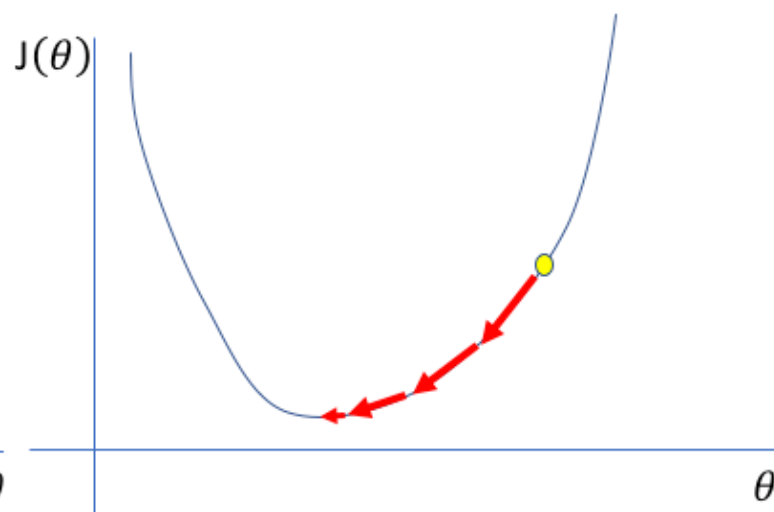


Too low



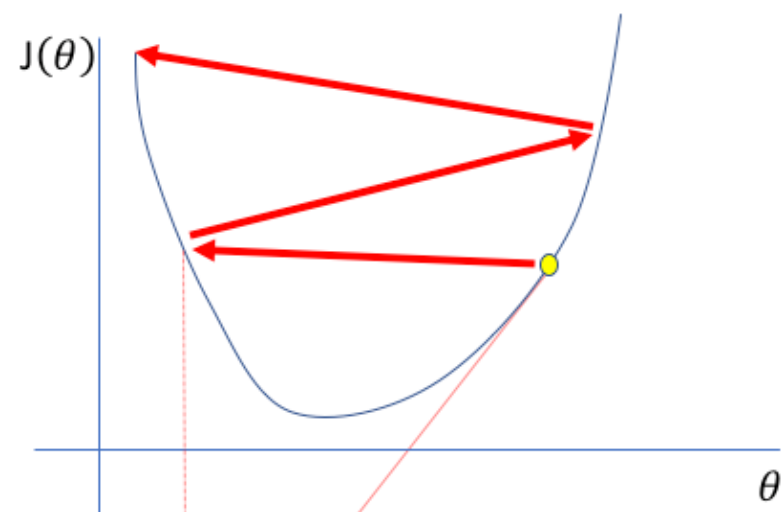
A small learning rate requires many updates before reaching the minimum point

Just right



The optimal learning rate swiftly reaches the minimum point

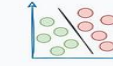
Too high



Too large of a learning rate causes drastic updates which lead to divergent behaviors

Model Loss Function

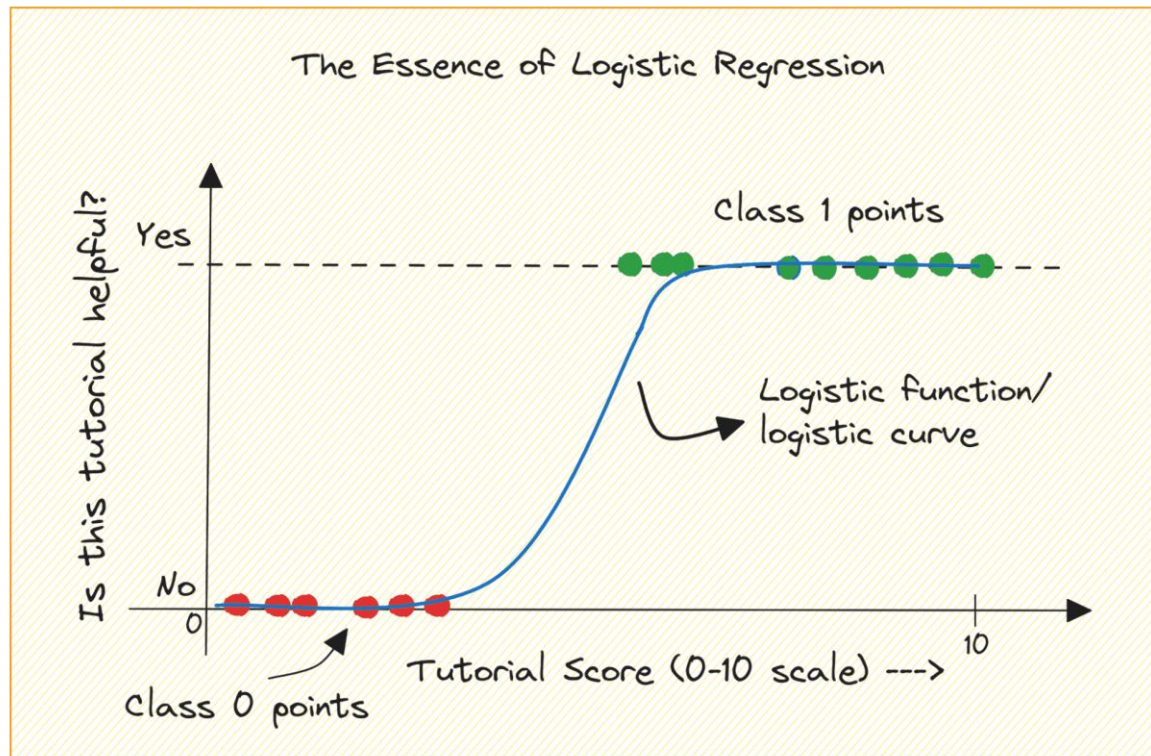
10 Most Common Loss Functions in Machine Learning



blog.DailyDoseofDS.com

Loss Function Name	Description	Function
Regression Loss Functions		
Mean Bias Error	Captures average bias in prediction. But is rarely used for training.	$\mathcal{L}_{MBE} = \frac{1}{N} \sum_{i=1}^N (y_i - f(x_i))$
Mean Absolute Error	Measures absolute average bias in prediction. Also called L1 Loss.	$\mathcal{L}_{MAE} = \frac{1}{N} \sum_{i=1}^N y_i - f(x_i) $
Mean Squared Error	Average squared distance between actual and predicted. Also called L2 Loss.	$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - f(x_i))^2$
Root Mean Squared Error	Square root of MSE. Loss and dependent variable have same units.	$\mathcal{L}_{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - f(x_i))^2}$
Huber Loss	A combination of MSE and MAE. It is parametric loss function.	$\mathcal{L}_{Huberloss} = \begin{cases} \frac{1}{2}(y_i - f(x_i))^2 & : y_i - f(x_i) \leq \delta \\ \delta(y_i - f(x_i) - \frac{1}{2}\delta) & : otherwise \end{cases}$
Log Cosh Loss	Similar to Huber Loss + non-parametric. But computationally expensive.	$\mathcal{L}_{LogCosh} = \frac{1}{N} \sum_{i=1}^N \log(\cosh(f(x_i) - y_i))$
Classification Loss Functions (Binary + Multi-class)		
Binary Cross Entropy (BCE)	Loss function for binary classification tasks.	$\mathcal{L}_{BCE} = \frac{1}{N} \sum_{i=1}^N y_i \cdot \log(p(x_i)) + (1 - y_i) \cdot \log(1 - p(x_i))$
Hinge Loss	Penalizes wrong and right (but less confident) predictions. Commonly used in SVMs.	$\mathcal{L}_{Hinge} = \max(0, 1 - (f(x) \cdot y))$
Cross Entropy Loss	Extension of BCE loss to multi-class classification.	$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \cdot \log(f(x_{ij}))$ N : samples; M : classes
KL Divergence	Minimizes the divergence between predicted and true probability distribution	$\mathcal{L}_{KL} = \sum_{i=1}^N y_i \cdot \log\left(\frac{y_i}{f(x_i)}\right)$

Logistic Regression



- **Logistic Regression** is a fundamental, supervised machine learning algorithm used primarily for **classification problems**, despite having "regression" in its name.
- Instead of predicting a continuous numerical value, it predicts the **probability** that a given input belongs to a specific, discrete class



Core Mathematical Concept

- Logistic regression builds upon the concept of linear regression. It takes a linear combination of input features and passes the result through a mathematical function to constrain the final output between 0 and 1.
- 1. The Linear Predictor
 - First, the algorithm calculates a raw score (z) based on the input features (X), their corresponding weights (w), and a bias/intercept (b):

$$z = w_1x_1 + w_2x_2 + \dots + w_mx_m + b$$

Core Mathematical Concept

- 2. The Sigmoid (Logistic) Function

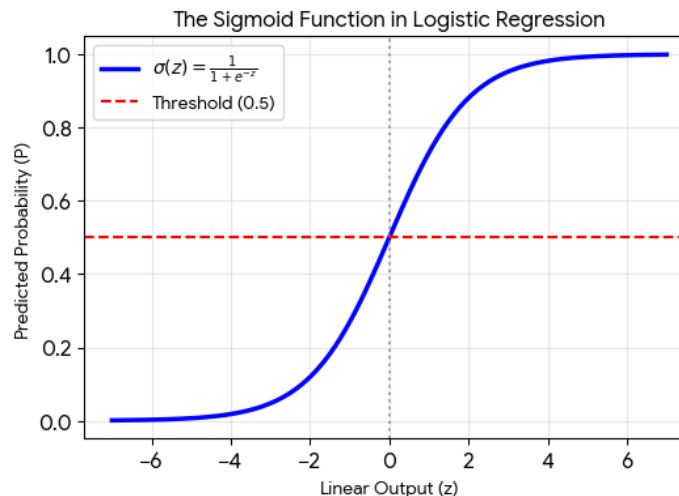
- Because z can be any real number from negative to positive infinity, it cannot directly represent a probability. To fix this, the algorithm applies the **Sigmoid Function**. This creates a distinct S-shaped curve that maps any value into a range strictly between 0 and 1:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

- 3. Log-Odds and Decision Boundary

- The output $\sigma(z)$ represents the probability $P(Y=1|X)$. To turn this probability into a definitive class (e.g., "Spam" or "Not Spam"), a **classification threshold** (usually 0.5) is applied

- If $P \geq 0.5$, predict Class 1.
- If $P < 0.5$, predict Class 0



LARGE LANGUAGE MODELS

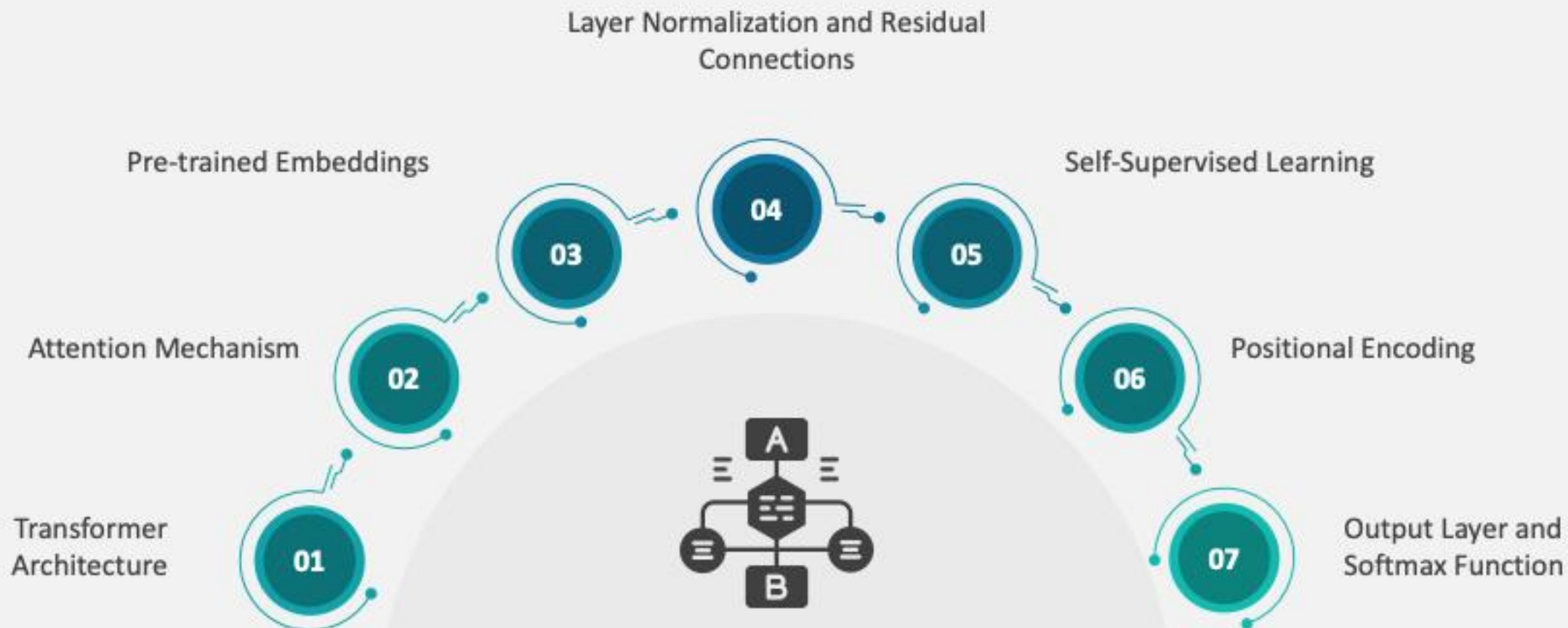
What is Large Language Model?

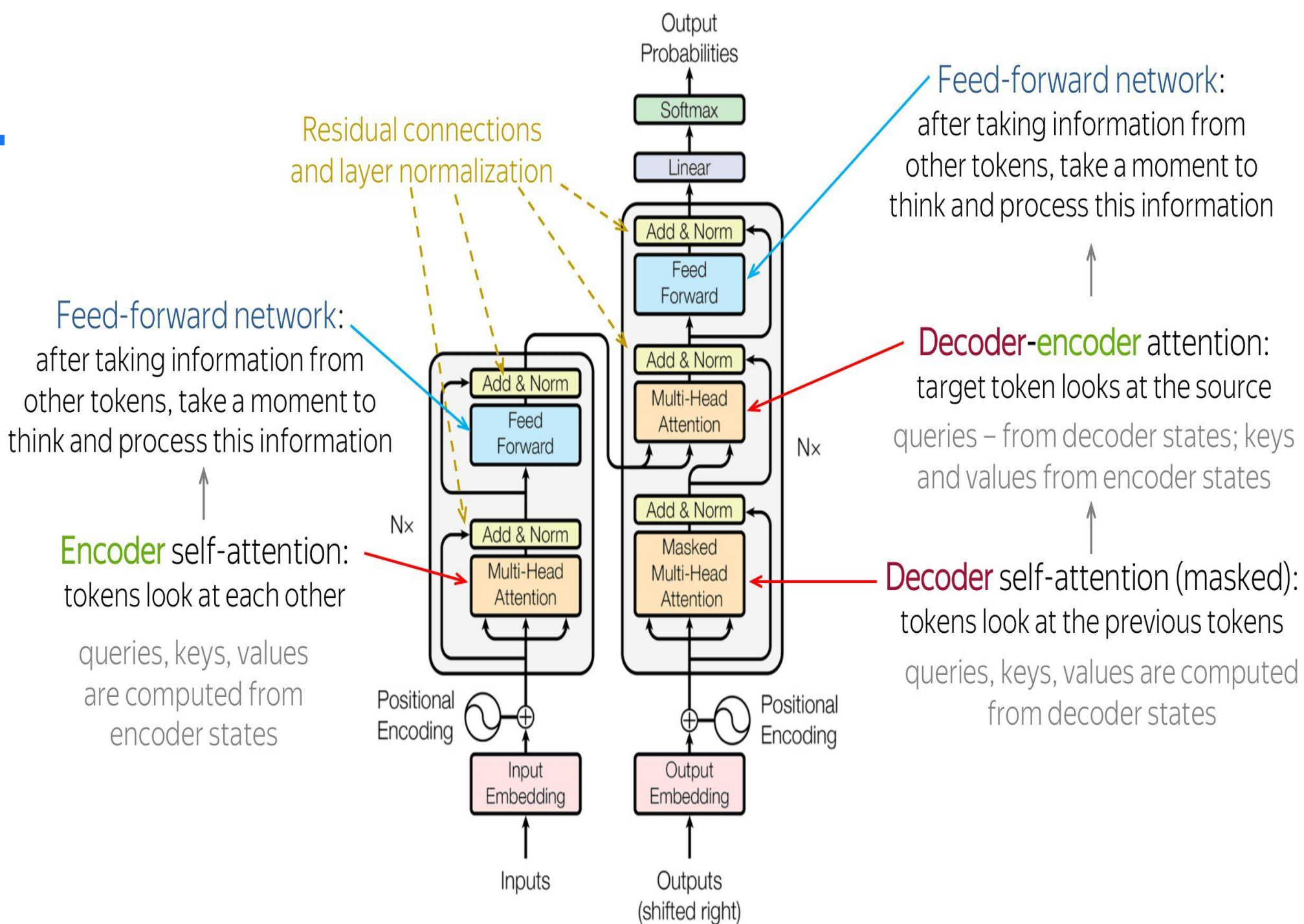
Large Language Models (LLMs) are artificial intelligence (AI) programs that use machine learning to generate and understand natural language. They are trained on massive amounts of data and use complex algorithms to understand the patterns and structure of language. Some popular large language models include LLM-style models and transformer models.



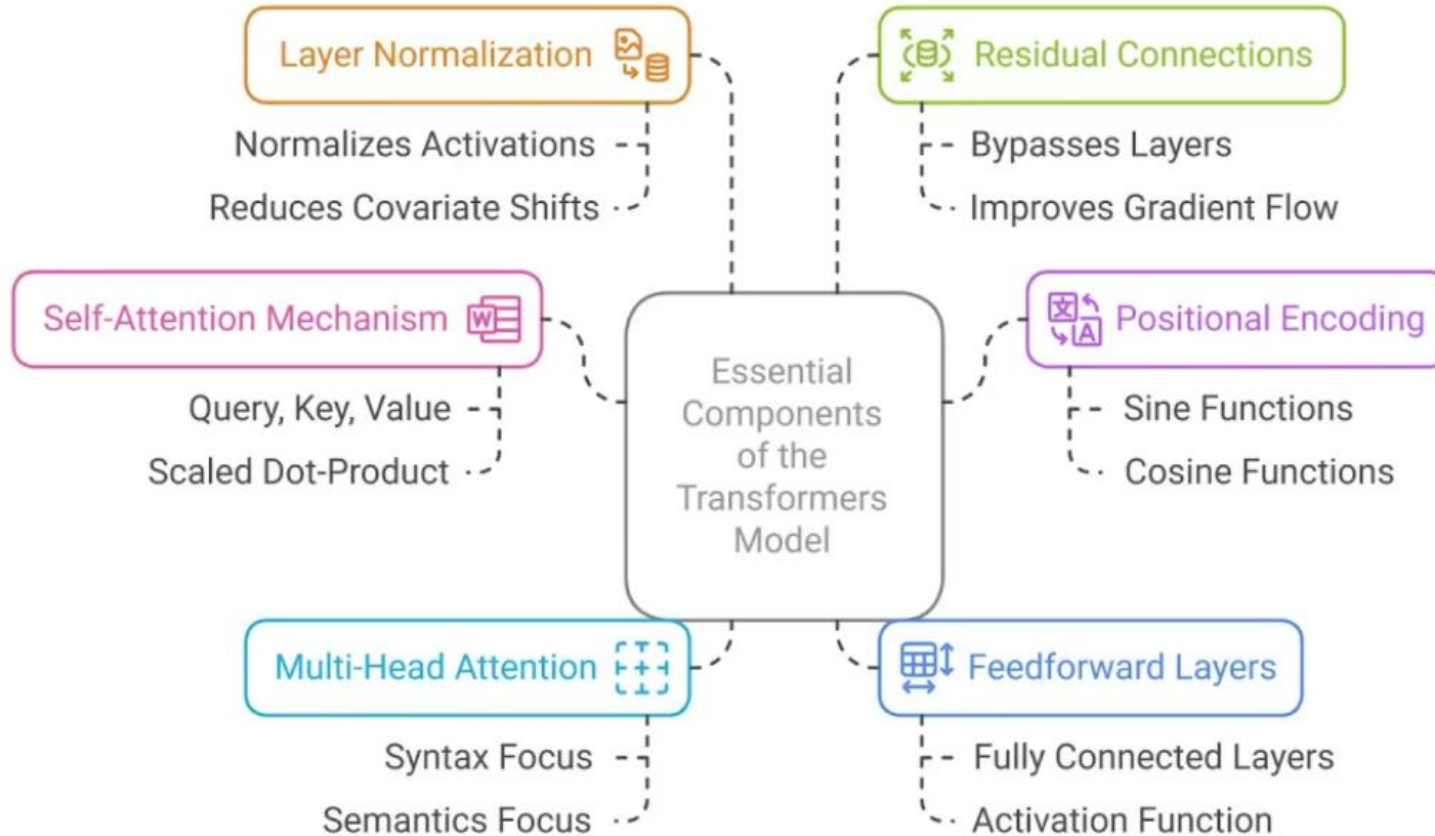
LARGE LANGUAGE MODELS

Key Components of LLM





Essential Components of the Transformers Model



[Transformer Explainer: LLM Transformer Model Visually Explained](#)

LARGE LANGUAGE MODELS

Challenges of LLM

Bias and
Fairness



Data
Privacy



Environmental
Impact



High
Complexity



Dependence on
Training Data



Lack of Contextual
Understanding



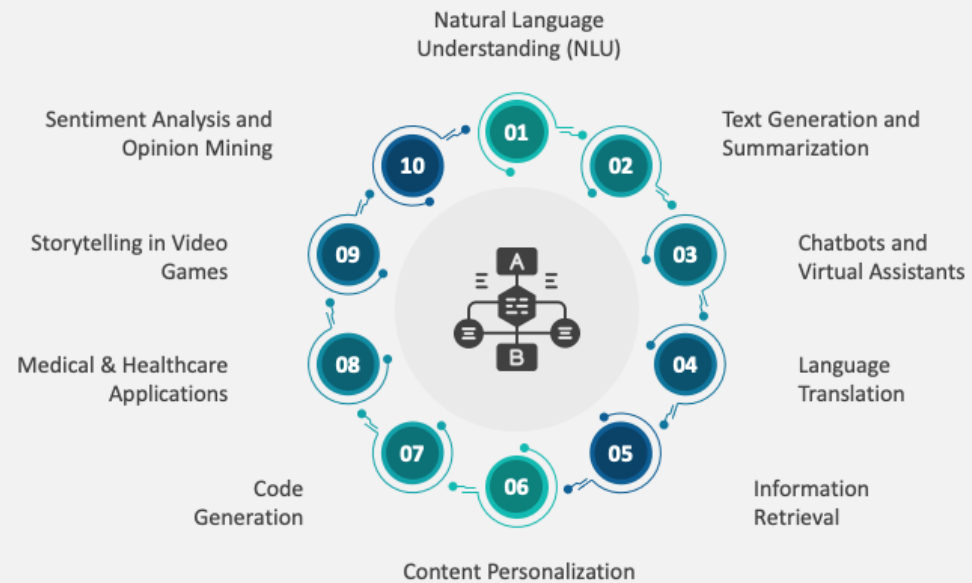
Ethical Use and
Misuse



Misinformation and
Manipulation

LARGE LANGUAGE MODELS

Applications of Large Language Models



Applications of Large Language Models



Best Large Language Models (LLMs) of 2024

A Comprehensive Comparison and Analysis



2024 has seen an explosion of LLMs with state-of-the-art capabilities. Here's a comparison of the best models based on performance, features, context length, pricing, and accessibility.

Top LLMs Comparison (2024)										
Model	Organization	Parameters (Approx.)	Context Length	Performance (Benchmarks)			Features / Strengths	Pricing (Per 1M Tokens)		Open Source
				MMLU	GPQA	HumanEval		Input	Output	
GPT-4o	OpenAI	~1.8T	128K	88.7	53.6	90.2	Multimodal (text, vision, audio) Best all-rounder, high reliability	\$5.00	\$15.00	✗
Claude 3.5 Sonnet	Anthropic	~Unknown	200K	88.0	59.4	92.0	Strong reasoning, long context Safer outputs	\$3.00	\$15.00	✗
Gemini 1.5 Pro	Google	~Unknown	1M	85.9	46.2	71.9	Massive context window Great for long documents	\$3.50	\$10.50	✗
Llama 3.1 70B	Meta	70B	128K	85.2	46.7	80.5	Open source, strong performance Great for customization	\$0.59*	\$0.79*	✓
Mistral Large 2	Mistral AI	123B	128K	81.2	39.8	81.1	High performance Cost-effective	\$2.00	\$6.00	✓
Qwen2 72B	Alibaba	72B	128K	84.1	44.0	80.3	Strong in coding & math Multilingual	\$0.50*	\$0.50*	✓
Command R+	Cohere	~Unknown	128K	83.0	42.1	70.0	Enterprise-ready Great for RAG & agents	\$3.00	\$15.00	✗
DBRX Instruct	Databricks	~132B	128K	80.1	-	-	Optimized for enterprise Strong tool use	Contact Sales	Contact Sales	✓

* Pricing for open-source models is estimated for self-hosted inference (infrastructure cost varies).

★ Key Takeaways

- ✓ GPT-4o leads in overall capability and multimodality.
- ✓ Claude 3.5 Sonnet excels in reasoning and safety.
- ✓ Gemini 1.5 Pro offers the largest context window (1M).
- ✓ Llama 3.1 70B, Mistral Large 2, and Qwen2 72B are excellent open-source choices.
- ✓ Choose based on your use case: performance, cost, context needs, and deployment preference.

📊 Benchmarks Explained

MMLU : Measures knowledge across 57 subjects
GPQA : Graduate-level Google-Proof Q&A
HumanEval : Coding ability (Python)
Higher score = better performance

📁 Use Cases

- Chatbots & Assistants
- Content Generation
- Coding & Dev Tools
- Research & Summarization
- Data Analysis & Insights
- Education & Tutoring
- Business Automation
- Creative Applications

🎯 How to Choose the Right LLM?

- 🏆 Best Overall Performance → GPT-4o
- 📄 Largest Context Window → Gemini 1.5 Pro
- 🔒 Best for Reasoning & Safety → Claude 3.5 Sonnet
- 🌐 Best Open-Source Overall → Llama 3.1 70B
- 💰 Best for Cost Efficiency → Mistral Large 2 / Qwen2 72B
- 🏢 Best for Enterprise → Command R+ / DBRX Instruct

Open Source vs Closed Source

Open Source		Closed Source
✓ Transparent		✓ Easy to use
✓ Customizable		✓ High reliability
✓ No vendor lock-in		✓ Better support
✓ Cost-effective (long term)		✓ Advanced features (out of the box)



What's Next?

LLMs are evolving rapidly with better reasoning, multimodality, longer context, lower costs, and more customization.

The best model is the one that fits your needs.



Pro Tip: Evaluate models on your own data and real use cases before committing.



Stay Updated: The LLM landscape changes fast. New models keep pushing the boundaries!